BEYOND REALITY

Evolution of VR and AR

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BEYOND REALITY

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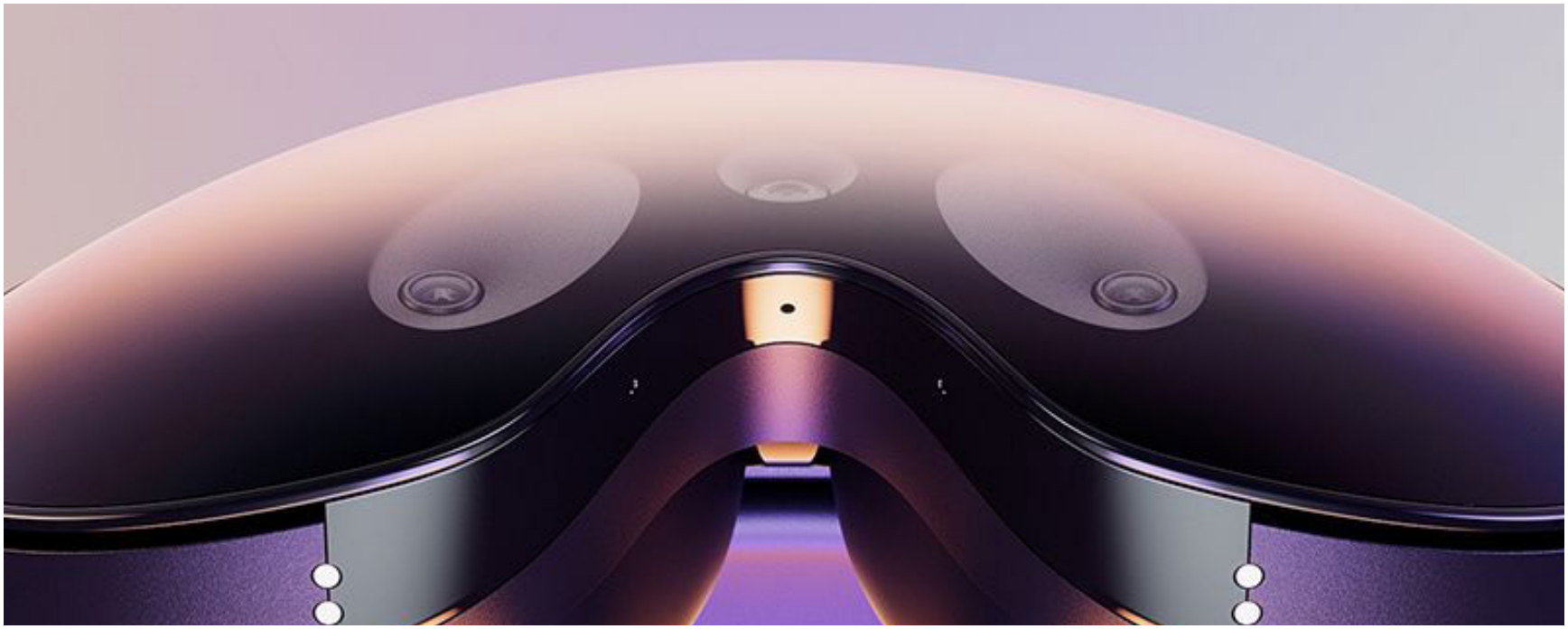

Author’s Note

Reality has always been more than what we see. From simple optical illusions to digital worlds that blend with our own, humans have constantly searched for new ways to experience what lies beyond the ordinary.

Beyond Reality is our attempt to explore that journey through six eras of immersion. We wanted to look beyond the inventions themselves and understand how each step brought us closer to the experiences we have today.

Through this magazine, we invite you to travel through the past, explore the present, and imagine what the future of augmented and virtual reality might hold.

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Introduction

Throughout history, media has evolved to bring us closer to new experiences, but AR and VR allow us to step directly inside them. By either overlaying digital features onto the real world i.e, Augmented Reality (AR) or transporting us into fully virtual environments i.e, Virtual Reality (VR), these technologies blur the line between the physical and digital, creating deeply immersive and interactive experiences.

Beyond gaming and entertainment, AR and VR are transforming fields like education, healthcare, architecture, and industry. By enabling students to explore complex concepts, professionals to practice high-stakes tasks, and creators to visualize designs before building them, these technologies are turning former science fiction into everyday tools for learning, working, and creating.

BEYOND REALITY explores this evolution through six generations of immersion, tracing the ideas, inventions, and breakthroughs that shaped the journey. From the earliest experiments to today's spatial and AI-driven experiences, this is a story of how technology continues to expand our perception of reality — and what might lie beyond it.

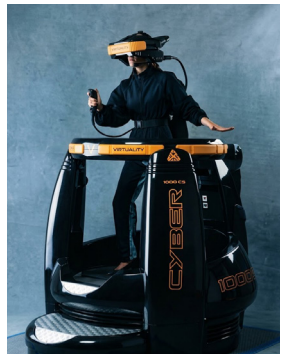
The next generation of AR and VR may move beyond screens and controllers towards experiences that respond naturally to our gestures, voices, movements, and surroundings. Virtual environments could adapt to our actions in real time, while digital objects become seamlessly integrated into the spaces we live, learn, and work in. As these technologies evolve, the boundary between the physical and digital worlds may become increasingly difficult to distinguish. However, this future also brings questions about privacy, accessibility, human connection, and our relationship with reality. The challenge is not simply to make digital worlds more realistic, but to make them meaningful, useful, and responsible. Ultimately, AR and VR are about expanding how we experience, understand, and imagine the world — creating possibilities that once existed only in our imagination.

STEREOSCOPE

Charles Wheatstone’s stereoscope showed that merging two slightly offset 2D images creates 3D depth, laying the foundation for modern virtual reality.

DATA GLOVE

The Data Glove used motion sensors to let users control virtual worlds through natural hand gestures.



VIRTUALITY ARCADE

Virtuality brought real-time 3D VR gaming to public arcades, introducing multiplayer experiences with headsets and motion controllers.

From 1838

T I M E

SENSORAMA

Morton Heilig’s Sensorama combined 3D visuals, stereo sound, vibrations, and scents to create a pioneering multi-sensory cinema experience.

SWORD OF DAMOCLES

The Sword of Damocles pioneered modern VR as the first head-mounted display with real-time head-tracking wireframe graphics.

VIRTUAL FIXTURES

Virtual Fixtures pioneered practical AR by overlaying digital visual cues onto physical reality to improve task precision.

INDUSTRIAL AR

Industrial AR overlaid digital instructions onto physical machinery to guide real-time assembly and maintenance.

MICROSOFT HOLOLENS

HoloLens merged digital holograms into physical space using gesture and voice controls for real-world applications.

META QUEST

Introduced mixed reality, combining immersive virtual environments with real-world interaction.

L I N E

To 2026

GOOGLE GLASS

Google Glass introduced hands-free smart glasses that projected digital notifications and navigation into the user's field of view.

POKÉMON GO

Pokémon GO brought AR to the masses by overlaying digital creatures onto real-world locations via smartphone GPS.

APPLE VISION PRO

Introduced spatial computing, enabling users to interact with digital content using eyes, hands and voice.

01

Mind Before Screens

Making the Unreal feel Real

Long before screens could build virtual worlds, people experimented with lenses, images and sensory illusions. The goal was to make us see more than what was physically there.

A vintage wooden stereoscope

The person using the Sensorama immersive viewing machine.

The first tricks of immersion

STEREOSCOPE

1838 - Two slightly different images, one for each eye. The brain combines them, creating the illusion of depth.

SENSORAMA

1950s - Morton Heilig engineered total immersion, blending sight, sound, motion, wind, and aroma beyond traditional cinema.

TELESPHERE MASK

1960 - An early head-mounted viewing concept that anticipated the shape of future VR headsets, combining stereoscopic images with sound.

An experience that could’nt change

These inventions could trick our senses, but their experiences were largely fixed. The images and sensations were prepared in avance, they could not create a new world that responded to us in real time.

Next: When computers began creating worlds of their own

02

Creating Virtual World

What If a Computer could build a World?

Virtual reality began attracting attention beyond research labs. Arcades, training simulators and experimental systems offered glimpses of digital worlds, while early augmented reality explored placing digital information over the physical world.

The person is wearing early head-mounted display systems

The girl wearing a head-mounted display and sensor-equipped gloves during a NASA virtual reality research experiment.

From pixels to presence

SWORD OF DAMOCLES

1968 - Ivan Sutherland and Bob Sproull developed an experimental head-mounted display that showed computer-generated graphics aligned with the viewer's position.

DATAGLOVE

1980s - Sensors tracked hand movements, giving users a way to control or interact with computer-generated environments through gestures.

NASA VIEW

Late 1980s - NASA researchers explored head-mounted displays, hand tracking and immersive interfaces for applications.

The person using a wired DataGlove to interact with a computer.

The world could respond

For the first time, computers could generate environments that changed with a user's movement or input. But the technology remained expensive, bulky and mostly confined to research labs and specialised settings.

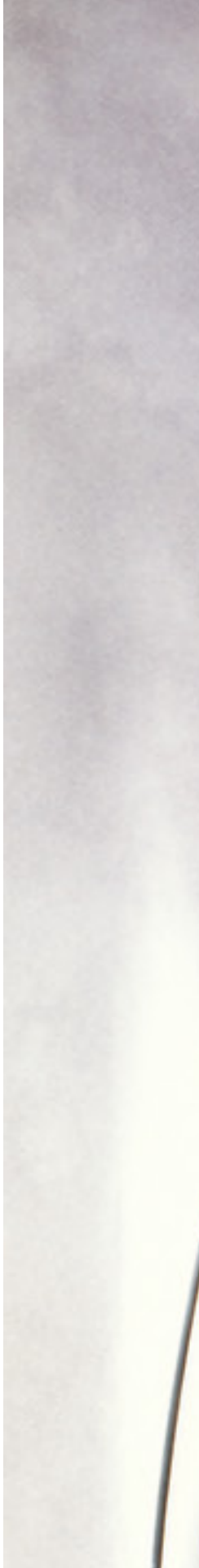
Next: VR and AR leave the lab and enter real-world experiments.

03

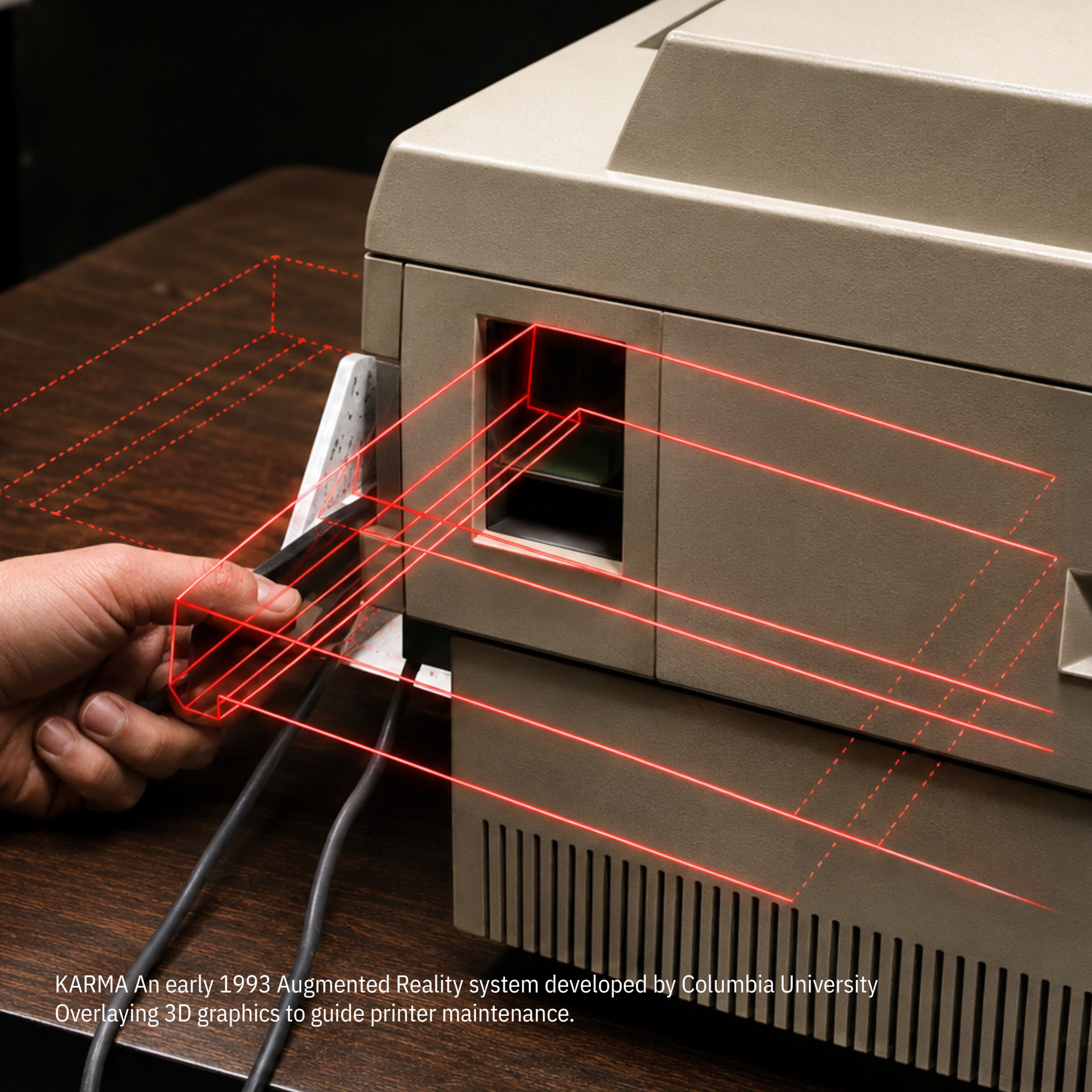
Emerging Realities

The Future Arrives. Almost!

Long before screens could build virtual worlds, people experimented with lenses, images and sensory illusions. The goal was to make us see more than what was physically there.



The person using a Virtuality VR gaming system to experience an immersive virtual environment.



KARMA An early 1993 Augmented Reality system developed by Columbia University
Overlaying 3D graphics to guide printer maintenance.

Two Worlds, new possibillties

VIRTUALITY ARCADES

1990s - Commercial VR arcade systems let people experience immersive 3D games using headsets and tracked controls.

VIRTUAL FIXTURES

1992 - Louis Rosenberg's system explored how virtual objects could be overlaid onto a user's view of the real world, helping demonstrate the potential of augmented reality.

NASA VIEW

1990s - Researchers explored displaying digital instructions and information over real equipment to support tasks such as assembly and maintenance.

The person wearing a head-mounted VR system and wired motion-tracking gloves, interacting with a virtual environment.

A great idea, a difficult experience

Immersive technology could entertain, train and assist people. But heavy equipment, high costs, limited graphics and complicated setups kept it out of everyday life.

Next: What happens when immersive technology meets the device already in our pockets?

04

The Touch Era

The World fits in your hand

Touchscreens and smartphones changed how people interacted with digital content. Cameras, apps and more powerful mobile devices made digital experiences easier to access. Affordable smartphone-based VR viewers also brought a first taste of virtual reality to a wider audience.

A close-up photograph of Oculus VR headset

A photograph of an HTC Vive virtual reality headset.

A screen becomes a window

GOOGLE GLASS

2012 - Unveiled in 2013, Google Glass introduced hands-free computing via a micro-prism display. Controlled by voice and touch, it established the UX foundation for modern smart glasses.

OCULUS RIFT DK1V

2013 - Researchers explored displaying digital instructions and information over real equipment to support tasks such as assembly and maintenance.

GOOGLE CARDBOARD

2014 - A low-cost cardboard viewer used a smartphone to create a simple VR experience, making experimentation more accessible.

A white PlayStation VR2 headset displayed with its controllers.

Accessible, but still screen-bound

Digital experiences had become portable and familiar. Yet most interaction still happened through a flat screen, and phone-based VR had limitations in tracking, comfort and immersion.

Next: Digital content begins to recognise the space around us.

05

Spatial Perception

When Digital met Physical

Better cameras, sensors and computer vision helped devices recognise surfaces, estimate depth and track movement. Digital objects could now appear anchored to real spaces instead of floating randomly on a screen.

Microsoft HoloLens displayed against a blue background.

A close-up photograph of Microsoft HoloLens 2, a mixed-reality smart glasses headset.

World becomes the interface

MICROSOFT HOLOLENS

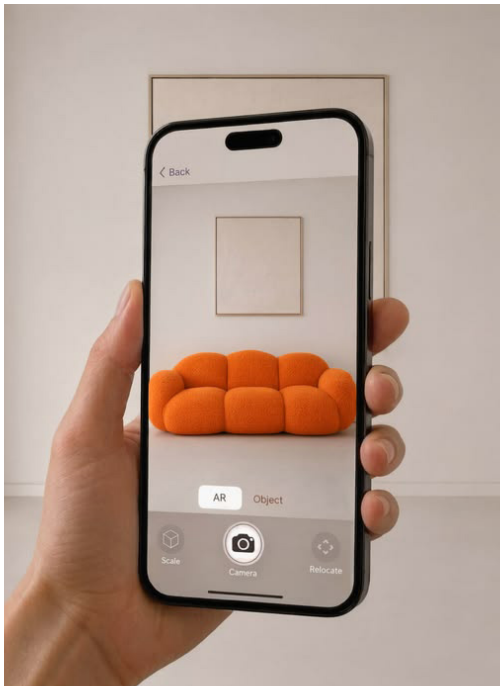
2016 - A mixed-reality headset placed digital holograms into the user's view of the physical environment, supporting applications beyond entertainment.

POKÉMON GO

2016 - Location-based gameplay brought digital characters into real-world places, introducing millions of people to a new relationship between physical location & digital experience.

AR TOOLS

2017 - Apple's ARKit and Google's ARCore helped developers build mobile AR experiences that understood surfaces and movement.



A digital Pokémon appears in the real-world environment through a smartphone camera, demonstrating Augmented Reality.

The space understands us

Digital content could now respond to the room, the surface and the user's movement. But many experiences still depended on specific devices, apps and carefully designed interactions.

Next: Spatial computing meets AI for more natural interactions.

06

Spatial Computing & AI

What if the Digital World worked around you?

Spatial computing brings digital content into our physical surroundings, while advances in AI create new ways to understand speech, objects and intent. Together, they are pushing experiences towards more natural ways of interacting with technology.

A photograph of Apple Vision Pro

Close up shots of Meta Quest 4

Beyond the screen

META QUEST

2019 - Headsets can show a live view of the surroundings and blend digital objects into that view, bringing virtual experiences closer to the physical world.

RAY-BAN AI

2023 - Introduced by Ray-Ban and Meta, these smart glasses integrated cameras, audio and AI into everyday eyewear, advancing hands-free wearable technology.

APPLE VISION PRO

2024 - Apple introduced a spatial-computing headset designed around digital windows, apps and 3D content placed within the user's environment.

The space understands us

We have moved from optical illusions to computer-generated worlds, portable screens and digital objects that share our physical space. Now, spatial computing and AI are exploring how these experiences might become more useful, personal and natural.

VR and AR in Everyday Life

Augmented Reality (AR) and Virtual Reality (VR) are gradually becoming part of our everyday lives, transforming how we interact with the world around us. From shopping and education to healthcare, entertainment and communication, these technologies are creating new ways to experience information. AR adds digital elements to our physical surroundings, allowing us to try products virtually, navigate unfamiliar places and interact with digital information in real time.

VR, on the other hand, takes us beyond the physical world into fully immersive digital environments. It allows students to explore historical places, doctors to practise medical procedures and users to experience games and virtual spaces. Together, AR and VR are making everyday experiences more interactive, engaging and immersive, turning what once seemed like science fiction into an increasingly familiar part of modern life.

What comes next?

The world becomes the Interface

The future of immersive technology may not depend on bulky headsets or handheld screens. Digital experiences could become lighter, more natural, and seamlessly integrated into the spaces around us.

Four possible developments

Invisible Interfaces — Lightweight glasses could replace some everyday screen interactions.

AI-Powered Realities — Intelligent digital assistants could adapt virtual environments to individual needs.

Holographic Communication — More immersive remote meetings could make distance feel less restrictive.

Smart Environments — Homes, classrooms, and workplaces could respond to gestures, voice, and spatial data.

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